

Active Inference Stream #010.1: "A variational approach to scripts" (2020)

1:23:06

hello everyone

and welcome to the active inference live stream

this is active inference live stream

10.1

and it is december 8th 2020 we are really excited to bring you a scripted or unscripted not sure maybe we'll figure it out

discussion on some exciting topics so let's

get right into it i will share my screen for the participants

welcome to the active inference lab everyone we are an experienced experiment and experience i guess in online team communication learning and practice related to active inference

you can find us at our website on twitter

at our email address our youtube channel or

public key based team and username this is a

recorded and an archived live stream so please provide us with feedback so that we can improve on our work

all backgrounds and perspectives are welcome here

and just as far as video etiquette for live stream goes

remember to mute if there's noise in your background raise your hand so that we can hear from everyone

on the stack and use respectful speech
behavior

also just a logistical note all the rest
of the meetings

for 2020 are going to be at 7 30 to 9
a.m

pst and today we are in 10.1

discussing a variational approach to
scripts next week at the same time
we're going to be having a follow-up
discussion 10.2

and then in the last two weeks of
december we're

going to be talking about the paper
sophisticated effective inference
and that will be on the december 22nd
and 29th

so contact us if you want to participate
in these discussions or

in 2021 or if you have an idea for
something that we could do

on an active stream today

in act in stream 10.1 we are going to go
through

introductions and warmups and then we'll
go to template one

we're going to be discussing the paper a
variational approach to scripts

and if you want some background context
you can check out active stream 10.0

we're going to talk today about the aims
and claims

or the hopes as they were and the
abstract and the roadmap for the paper
thankfully with one of the authors on
board and then we'll talk about some
different kinds of scripts

talk about scripts and their uses ask a

few questions related to that
and then close with looking through the
figures and just discussing what they
might mean
and next week in 10.2 we're going to be
on this same paper
so save and submit your questions or if
it's after 10.2 just write them in the
comment section
and get in touch if you want to
participate
for the introductions and warmups let's
introduce ourselves just by giving a
short introduction or check-in
and then we'll pass it to somebody who
hasn't spoken yet
so i'll start i'm daniel i'm
excited for this conversation and i will
pass it to
our first time visitor mao
hi so yes my name is mao alba
i'm one of the authors on the paper
along with axel constance maxwell
ramstead and karl fristen
i'm a first year phd student in quebec
and i'm working in cognitive computing
uh to try to establish how we can
formalize
epistemic communities through active
inference
cool then we'll pass to another first
time guest rj
sure uh so my name is rj corday uh i'm
in new york
and i'm currently researching
organizational dynamics intelligence
analysis and narrative warfare keep it
short

about sasha

thanks hi i'm sasha i'm based out of
california

and i'm also a graduate student studying
developmental neuroscience

with that i'll pass it on to blue

blue muted we can hear you

awesome thank you i'm blue knight i'm
from new mexico

and i am an independent research
consultant and i'm currently working on
agent-based modeling and a bunch of
other random things

i'll pass it to shannon

hey i'm shannon i'm typically based out
of california

i'm also a phd student i'm interested in
music neuroscience or i work on music
neuroscience

and i'm interested in

music and social interactions so i was
really excited

for that social side of this paper and

i'll pass it to yvonne

hello my name is ivan i'm based in
moscow and

i'm interested in active inference as it
goes with the systems engineering

and i'm very excited about this paper i
found it

uh very interesting cool

thanks everyone for coming out and for
everybody who's watching it

live and replay so for the warm-up
questions

section everybody is welcome to throw in
a thought

i'll put up the uh first two questions

and just raise your hand so that we can
hear from anyone the first question
is what interests you or excites you
about studying scripts
so kind of what drew you to this paper
whether from the active inference side
or from the social sciences side
and then what's something that you
thought about after

reading the paper so while people are
raising their hand i'll start

i think what interested me in uh
studying scripts

was just addressing all of these
different social contexts with one
integrated framework that's kind of an
appealing idea

mao

well yeah and i think it's also it's
sort of a road map that links
internal and external manifestations
of our general understanding of the
world and
systems dynamics that get um
brought together
through these islands of negentropy that
become these scripts

if that makes any sense cool

rj um so

uh after i read the paper what i ended
up thinking about

is what what do the conditions look like
for threshold events for changing
scripts um

that's a really interesting to me and
then um

what kind of like what what would the
metadata be and this is something we had

discussed

uh before is like what would the
metadata be if we wanted to catalog
the scripts that are that are operating
in a system

cool and that um bringing up that
negentropic

idea the sort of organizing principle of
scripts

that's pretty interesting too because uh
in combining what both of you just said
how do we change a script from one
island of

stability or organizational attractor to
another organizational attractor

if we go from one organizing principle
maybe we don't like some aspects of it

but if we just go from order to
pure disorder that's not doing the
society of service so we want to move
from order to order

ideally through order but how is that
actually going to be

uh tracked or designed and then just
the last up shannon go ahead yeah that's
actually exactly the one question i had
written on my paper

um are is what age rj and um

daniel mentioned like what

can active inference help us to identify

like when a critical transition will
happen or when

we'll switch scripts um it's really easy
in

it's really easy i say but in in music

we have scripts

and it's either written on a piece of
paper or it's

the cultural transmission of how you
perform in this genre
of music alone or with others and
as long as you know that that's the
script that you're starting from you
know how the social interaction will
transpire
and then if you want to do an experiment
you're like you know
that's the constraints that you have and
you can pick out the interesting
features
that you want to empirically measure
like whether it's body sway or the
actual music that's being played
but um in the example that runs through
the paper of flirting
you don't necessarily know the goal of
the interaction
if you're observing it or the goal of
the interaction if you're in it
because it's not codified in like this
very formal or traditional framework in
the same way that
like a musical interaction is so
identifying the threshold or identifying
how those transitions happen
would be very different in those two
types of interactions
cool blue
so i was excited to really learn more
about um
how scripts can interact in a
hierarchical way
like just the paper kind of mentioned it
but didn't really unpack it uh
thoroughly and i just would like to
really kind of dive into that

yeah and the music and the cultural side
it's uh maybe with a musical ensemble
they could agree to play a different
song within the same style or agree to
play a different style
and then we could ask the same with
culture but it's a little bit more
challenging to
simply adopt a different cultural regime
but maybe within a culture there's a lot
of latitude so there's this question of
nesting and
horizontal interactions and about style
controlled novelty so i think there's a
lot of cool
ideas we'll be bringing up any other
comments here on these warm-up questions
mal
up you're me too
sorry so yeah that's the idea that's the
goal the um
eventually measuring where the threshold
happens um
we have another paper that's being
written and that's
trying to identify how scripts
tend to become overlapping through in
interaction in society and when they
become overlapping they
they become analogical so um the scripts
of femininity and masculinity are not as
segregated as they used to be for
instance i'm wearing no
nose sleeves
in these 1800s but would definitely have
been acceptable for men
nowadays it's not a problem anymore both
men and women can wear it so these

these scripts have tended to overlap and
we're seeing that because
the scripts overlap they become a little
bit meaningless because they're not
allowing us to predict anything
so there's going to be to be a point
which we're kind of sensing already
where the concept of gender as a strict
binary
is just breaking apart and we're seeing
new
uh new concepts emerge new new
possibilities so this
this is where the threshold can be
measured and i think we can definitely
formalize that
as far as codification um
art in general is just a series of codes
that once you have you get to break it's
just it's become this sort of script in
art where
you have to break the script you you
have to use what we know
in order to continually create novelty
but
think of all of the hollywood movies
that depict flirting
it's pretty much the same like there's
there's a few scripts that are possible
but if you look at it abstractly
it's basically you know you don't you
don't flirt necessarily
uh with someone you have zero intention
of ever ever becoming with
in hollywood movies or you don't flirt
with someone by
hitting them on the head with a flounder
you know like

like there's some things that you know
you're supposed to do and some things
you know you're not supposed to do
and the strictness depends on
how wide one vision of the script
becomes
thanks rj i just uh
something i picked up from what you were
saying there uh on um
on flirting in in in cinema like
i feel like a lot of the comedies i know
of are actually about
script breaking so even even like
outside flirting like something like uh
there's there's a like common category
like in in in the uk it's peep show in
the us it's
always sunny um where it's most of the
show is just about
failing to adapt to cultural scripts so
i just want to know like i feel like
there's there's some element there where
we actually like contrast like exactly
what it is we're not supposed to do as
well
yep there's the exception that proves
the rule and then there's the
the chess master who within the
structure of the rules
makes an innovation that appears in the
short term to be breaking the rules
and then there's this element of the
media which is showing us for example
maybe flirting scenes with things that
wouldn't work in real life
uh whether for anyone or for subset of
people but
often including very for example bold

maneuvers

they can't uh convey in the hollywood
reality meeting somebody getting to know
them over years

not as exciting not as a clear

foreshadowing so

maybe there are sort of rule or script
breaking things

but then when we see that representation
and maybe even starting to introduce
some of these

active ideas we see them in our sensory
states

and therefore we perceive them as likely
or plausible and then we come to embody
these beliefs about what is likely or
possible

or uh preferable in the world and we see
people

imitating behavior they see in the media
so that can be used for

good and for bad sasha

yeah this kind of loops back to the
question that i had after reading the
paper which is

um how do individuals convey
um their

true state or um their

you know unique perspective on the
script while conforming to the social
script

because um if we all acted like we were
in a rom-com

we would have different lives and
relationships but clearly we all have
very unique

um and uh personalized

um versions of that

and i don't think any of us live in a
rom-com but
you know please correct me if i'm wrong
yeah that's kind of the fun it's like
asking what's the genre what scripts
will my scripts uh or what genre will my
scripts be drawn from am i in a tragedy
am i in a war movie am i in a rom-com so
that's kind of
fun it's like what genre would your
theme music be what kinds of conversants
are you gonna
be interacting with um mao did you have
your hand raised and then aren't you
yeah
yeah go ahead so first thing is um
i think you mentioned the the
oppositeness and that's a really
interesting idea like the
that's also how new scripts emerge is
that you have these
these bubbles of counter discourse that
come
and that's either embody um embody
portray something that is real in
society but
often invisibilized or something that
um wants to push back on something that
feels oppressive to a large swaths of
society
so for instance one great study that i
saw i'm sorry about i i don't have the
name of it right now but i'll send it to
whoever wants it
it was about the portrayal of abject in
uk society and the point was
there's all these mako versions that
show you

first all these women how they are not
supposed to be so they take a woman
and they portray her as very ugly often
um lower class like they give her all
the qualities you're not supposed to
have and then they
take her through this transition and
they make over her life and suddenly at
the end of the show
she's happy her life is great her
children love her her husband loves her
again
it's great and it's like this this
movement from the abject that you're not
supposed to embody to the
to the ideal that you are supposed to
embody and on the second note
what you said i think sasha is that's
exactly right but the thing is
you are not the person in the rom-com
you do not perfectly embody what
she is even if they try to make her
relatable
you are connected through these tendrils
of semantics to a variety of things
that make you take what is possible and
adapt
all of these possible scripts into this
new
version of adapted contextual
anchored script which is why there is
variety as well
cool rj um i was gonna say
uh mao covered a lot of what i was gonna
say actually so we're
we're fairly saying i was just gonna say
like that um
i like actually i like the reference to

like makeover shows i was gonna i was
gonna talk about um
a very opposite end uh but
but uh greek art and stories where where
it's
it's very often and this is where like i
think joseph campbell
brings up the hero's journey um where
it's from some state
which is object negative not where we
want to be to
some other state none of us actually
live in the rom-com like but but the the
ideal being presented
um isn't something we're supposed to
mirror necessarily but
something that kind of acts as a strange
attractor um but
yeah but anyway mal covered most of what
i was gonna say there yeah rj
i i wrote down literally hiro's journey
because the makeover show if it's weight
loss or
about um norms for genders or house
makeover
or chefs it's always this hero's journey
and it's a scripted
hero's journey literally designed
journey and then the interesting part is
the policies
that they undertake thinking about
active those policies are not
our affordances they have a team and
they have a camera person and they have
a
coach so then it's like showing you
the first point and the last point on
the trajectory

and saying well now you figure out what policy but the only example that you've observed the only example of what's a plausible affordance is something that is defined as being inaccessible to basically everyone so that's just like such a twisted way to play with how people actually improve themselves yet mao and then rj yeah so that's that's exactly right you you touched on a very important point is that take each script as though it was its own little like market blanket like it's not right but imagine um there's some parts of the scripts that will interact with one another and then there's these like blankets that move a little slower and through the blankets you can go to another part of the script and you can do this at all the levers so this was touching on the idea that someone else brought earlier that there's layers of scripts and that's absolutely true you can break down every script into smaller smaller scripts what defines a script is its edges so entering the script and leaving the script what happens in between can also be scripted but that's at a lower level right so for me to for me to speak here i had to go through a series of things but what really matters is that

i'm here now and that this will be
recorded but
um getting here it doesn't really matter
how right unless i'm coming from it
as a as an academic so i had a series of
steps that i did to get there as an
academic
the the way that you bring these scripts
together your
vectors from where you're coming from
define
also what kinds of policies you'll have
available to you
inside a given script cool rj and then
we'll continue
uh yeah uh oh also um real quick i like
that that's that's quotable
what defines the script is its edges um
but uh
i just wanted to add real quick uh that
it's probably
actually dangerous to optimize too much
to say like the rom-com
or the ideal like we don't actually want
to copy it exactly right like i feel
like maybe maybe
uh um there's there's something there
where
where we're supposed to actually mutate
the scripts we're being presented
over time and that like if we actually
optimize them exactly like a perfect
mirror then we actually get instability
but anyway
cool well i hope if you're listening
whether from the active inference side
hearing about scripts for the first time
or on the script side hearing about

active inference you're starting to see
where we are all getting excited about
at this intersection the paper that we
read today
is a variational approach to scripts and
the aims claims and hope of the paper
were addressed
in 10.0 so feel free to pause this video
and recursively watch the other one if
you want
but mao since you're with us what would
you say in your own words were the
aims and claims or the intentions of the
paper
so the goal was to bring together
fields that had different strengths the
active inference right now is
trying to move towards the the social
dynamics
it's doing a great job but there's still
some way to go
and in order to do so it doesn't have to
reinvent the wheel
on the other hand social sciences have
done
a lot of theorizing but it's it's
difficult to
assume causality it's difficult to see
them the true
mechanism of things without either doing
a lot of
qualitative work which itself has some
of the
the author's bias or specifically his
own expertise into it which takes a lot
of time
and it's the the kinds of
epistemic value you can draw from a lot

of these studies
is slightly lower and considered lower
by all of the formal fields so if we can
bring together
all of these fields offer a wonderful
tool to the social sciences we can bring
it all together and
bridge gaps in the in the communication
of
interdisciplinarity awesome
any comments on here but if anyone's
raising their hand good
and that reminds me a lot of the sort of
stem
the science tech engineering math
centrism
or bias it's tied up with reductionism
and a lot of other ideas
but then we see initiatives like steam
like throwing in the a for art or adding
in
crafts or maker diy all of these
different
things that are at the edges just like
you brought up at the fringe of the stem
narrative or at the fringe of the
accepted mainstream narrative
and it's people who are making a value
proposition
for some of these areas that haven't
been included in the
stem conversation so to speak and
there's various ways that this can
represent itself complexity science or
interdisciplinary work
project-based learning interdisciplinary
teams
this is really also why it's so

important to have different people's
perspective
on basically everything because this is
the shape that we enclose as a group
is everyone's perspective and that
includes our training all these
different
aspects so it's not just like a
unilateral uh
learning relationship where the
humanities have so much to learn from
machine learning scientists yes they do
and it's also
bi-directional so what does the
bi-directional
nexus look like and how are we going to
make that nexus
really a clean and accessible location
instead of just making it a a slide like
from one part of a playground to another
all right also in the abstract we uh
read that in 10.0 but mao again maybe
what would you
summarize this or i'm also just curious
like when you present it to different
contexts or audiences
so when you're in front of audience a
versus b let's say
what are two different ways that you
talk about what you're doing
um i i mean i tend to present it pretty
straightforwardly to anyone because it's
it's very simple
it's essentially just saying that um
we behave in ways that are predictable
because
our cognition about the world and our
position in the world is

also predictable through a bayesian
framework
most of our cognition is not perfect but
been that definitely basin and so
the beauty of active inference is that
it gives us a way to model
the different ways in which our
cognition can be slightly biased
or let's say not
classically logical but definitely
following
a gradient of optimization towards
 x and this x is like minimization of
free energy so
through the scripts that are both um
let's say simple and
more formal so scripts that are
weaker so semantic fields and stronger
so behavioral links between some of the
semantic elements
we can essentially predict the behavior
and the thought process
of anyone as long as we have information
about
their position in the world
cool and if i could just build a little
bit on that
it's kind of like the scripts are
archetypes or
patterns that play out on landscapes and
that's kind of like water
on a landscape and the landscape in this
case is like a landscape of affordances
a field of affordances
that's things that we've talked about
but it's also potentially
quantitatively a landscape of
free energy minimization or there are

entities that we can model as if they're
doing optimization
in a variational bayesian framework on
some
landscape that we're also inferring
about as investigators
and then just like water on the
landscape there might be
a marsh where there's a lot of ways to
go and it's not quite clear it's very
flat
there might be other parts that are like
the grand canyon where if you're
in the canyon you're totally in the
canyon and if you're not you're totally
not
and so that would be like a very strong
script in the grand canyon and then a
marsh or like a flood plain
would be like a weak script so that's
one dimension and then the other
dimension which we're going to come back
to
is this internalist versus externalist
so we'll get to that in a second but
that's another dimension
about how active inference can integrate
scripts
in 10.0 we also talked about the roadmap
and it has again the structure that we
know and love
where you introduce another field here
script theory in section two
and then activate active inference in
section three
and then turn in the second part of the
paper to
an integrative discussion so if you have

any comments now about the structuring
of this
paper feel free
um so this is a structuring this very
natural branstad
that's this thing um and i think
i think it works really well because
we're still at a point where a lot of
people don't necessarily understand what
we mean when we talk about active
inference
it's it it doesn't feel that esoteric
but when you read some of the papers
about it some of the language is a bit
complicated there's a lot of equations
so people get like
okay no this is not for me whereas the
the the main idea is really simple
there's an attractor and phenotypes or
people or
entities tend to revolve as
around the attractor in their states the
attractor moves a little bit so that the
states are not
constantly at equilibrium that's really
that's really
the extent of what you have to
understand to go through active
inference
cool yep there's so many uh aspects to
what you just
added there it's always important to
provide little mini on-ramp
in the paper itself to active inference
without perhaps going into all the
details
and also it's important for the
community to be formalizing and

developing

the real frameworks that we can refer
out to

so that we can have a short version for
some papers like this and then have a
really rich version that people know
they can always come back to

let's talk about these two dimensions of
script variability so here is just one
slide

but this is people's life slash career
slash

decade long literature fiasco's back and
forths

it's great times and we're just gonna
lay all those

roughly out on this two by two grid
and there's some images that kind of
reflect a few of those dimensions or a
few of those

points in the phase space here but now
just

you know so much more about scripts than
us um

where are some examples of some of
scripts that people

might recognize or where has this been
an especially contentious debate

or is there a difference between field a
and field b

like uh disagreement that stems from one
having an externalist perspective and
one having an internalist perspective
well it's it's not some emotion and
disagreement it's just that

depending on what your paradigm is
you'll tend to focus on externalism
and another paradigm you're tend to

focus on internalism like for instance
behaviorism could never have gone to an
internalist perspective because it
literally
tries to evacuate anything that happens
inside cognition it's like
what's the input and the output so this
would be fundamentally externalist
um social studies will tend to
to move through externalist perspectives
because they don't necessarily also have
access to
what the perception from the individual
is of a certain of a certain fact all
you have is
movements of demographics and and how
these movies
um manifest in the world so of course
it's never as simple as that there's
always
internal elements and external
misconceptions and external
misconception
elements and internal misconceptions but
it's it's where you tend to focus
in our paper we're showing that active
inference allows us to bring both
together because a generative model
is a model of you and the world
and the beauty of this is through the
work of axel constant
and uh samurai vessel we can see that
the world
also has a model of view so we have
this dual connection that's possible
through internalism and externalism
a good example of an external script is
institutions

uh you you you have to go through
certain
um passages in order to get something
you have to have a passport to go to
another country
and in a an internalist script is well
how do i want to dress up for something
tonight
do i know what to wear am i going to
wear something like
it's it's probable that i'll get in the
club if i wear xyz
and there's some clubs that would stop
me from going into the club if i don't
wear the proper clothes this is
both an internal script that i have as
an expectation
of what the club will allow me to do
given xyz
the weak script is we have some clusters
of categories so um i know that
some colors go with some things i know
that some words are appropriate in some
places
and some words are not appropriate in
other places but i don't necessarily
know the order of the words or i don't
necessarily know the order of the colors
or
like i can wear pink up here and yellow
down here or
yellow and pink there's no necessary
order or
structure to what i have to do so long
as i stay within these clusters
a strong script is when you go to
a wedding there's a series of events
that will happen only one person wears

white et cetera et cetera so
it's way stricter as to where each
element of the semantic clusters
happen to fall both in time and in space
cool thanks for that rj
uh thank you for that explanation uh
because i i
when i first read the paper i was
thinking that it's not so much
disagreement as much as it is uh the way
that the priors that different fields
are using when they're approaching the
subject
and the subjects that they're studying
so like when i thought about it was like
externalist week maybe is like io
psychology
where it's a strong function of the the
organizational culture it's much
but and then maybe extraordinary
externalist strong
um would be closer to maybe maybe in the
realm of geopolitics where it's looking
at like
procedures after after regime changes so
yeah
yep and also the way um i've thought
about this like
internalist and externalist is that
there's very little disagreement on what
is
for example people can say yes there's a
virus particle that
consists of lipids and protein and their
society and people are moving around
but then somebody who writes a review
paper from the perspective of a protein
biologist

they're going to focus on the proteins
and so it shouldn't be too surprising
that because of their regime of
attention
and their knowledge affordances they're
going to recommend
some type of molecular binding agent
whereas somebody who studies it from the
sociological perspective knows that the
proteins exist
but then they're going to highlight
different features and therefore going
to make different recommendations it
would just be
off-key if somebody wrote a sociological
review paper and then
went into a conclusion about a protein
binding molecule
so that's something that we need to
really understand that we're all at the
same table
but it's our different perspectives that
can either divide or
integrate us mal
yes so what you're bringing is really
important is the silence of causal links
so
the beauty of active inference is that
it allows us to draw
um causal chains right or at least what
we believe is the inference in
regard to the causal chains where you
come from
defines what is salient in order to get
to a goal
so that's why scripts are not just like
a point here
and you get there however you want right

that's that would be a very very very
weak

version of the script and anyone who is
in the market blanket of where anyone
tries to get to this point that's that's
a very weak script but a strong script
is well in order to get to this point
there's

x element that is salient um as you said
earlier if

if someone comes from a different field
and tells you no no the important causal
chain

is through externalist things it doesn't
matter what's inside your mind that's
behaviorism right they would say
it doesn't matter what you think so what
happens is if you don't do it you won't
go through

and the internalists will be well it
doesn't matter if you go through
what matters is how you act into the
world

but um you're ryan pointing out that if
you're

we're all at the same table we have to
realize that

there both of these things are kind of
true what really matters is how they
causally interact to get to

um a final product a final point

great point and here's something to add
on that is within active inference

we talk about causal inference uh the
latent causes in the world as

are hypotheses about the system and so
if one person hypothesizes that there's
a causal link and somebody else says i

think it's not as important as you think
it is or i think something else is more
important

then what we can say is instead of
getting into an inference

duel we can get into an active inference
collaboration

where we say great well given your
hypothesis about the causal network and
mine

what's the optimal experiment that we
can do

and optimal experimentation entails an
ethical structure

nobody says well my optimal experiment
is that we're gonna uh move a thousand
people to mars and we're gonna test them
with this

device that's not it's not tractable and
it's not within our affordances and it's
not ethical

so then we can say that's actually not
in the distribution of experiments we're
going to consider

gonna download that family of
experiments and so we can come to the
table

and ask what is going to be the most
informative experiment

that we can collaborate on specifically
to reduce our uncertainty

about our shared world our shared niche
interesting stuff mao that's exactly

right like we can come up with new
models of governance through this by
understanding that it doesn't have to be
a zero-sum game

it can literally be a productive game

where we're just
optimizing through the the world of
possibles that overlap
and that are meaningful for everyone
involved so
if you could model i'm being
i'm being a little um optimistic here
but if you could model
everyone's scripts and we could model a
shared
understanding of what would be a um
a desired state of the world which would
be like rg
we have this g then we can find the
policies that optimize
for um our our sets of constraints
that's just that's just beautiful right
there
awesome shannon i have
um the sort of same question i brought
up earlier
about in music you know what the goal is
you know what this um
goal state is that you want to optimize
towards in
social interactions once you categorize
them you know kind of what the goal
state
should be but if you don't know the gold
state so you just have
a couple different social interactions
maybe you filmed them
and and and you want to figure out can
active inference if you if you do
like an active inference model help you
figure out what kind of
social interaction the people in this
video recording are engaging in

oh okay now go for that so yeah so
there's two things to this the first is
because you're embedded in a culture
that has a set of narratives you
actually know
what the what the goal is of given
social interactions
whether it's implicit or explicit you
enter into
into certain social interactions because
you wish to enter in the goal or at
least
some polysemi of goals
the second thing is if you are not sure
you make inferences
about what that goal might be and then
you test those inferences by trying to
you know optimize for a goal you might
want
or you um you check if your prediction
was right and then you update so like
it's it's this push and pull that allows
you to get more information so
um disambiguating reality is something
that
is shown in active inference to be
fundamentally valuable so
me figuring out whether or not you're
flirting with me right now
can be a goal in and of itself even if i
don't care like i don't want to flirt
and i don't care that you're flirting
but i wanna i wanna know though i wanna
are you flirting is this an
accurate model of the world that i have
so that's one thing that active
inference allows us to
model in human behavior the second thing

is
because we we have scripts that
lead to certain pathways so certain
policies policies lead to goals
we can figure out what kind of policy
you would sort of be engaged in and
infer a potential goal you're leading
towards
if we're wrong eventually the model will
update
so we can see where are the attractors
in certain behaviors
which kind of behaviors tend to cluster
towards something and based on this we
could infer if we didn't already
know what is the social goal of a
specific action
shannon yeah awesome the
second part of your of of your answer um
there's been a lot of like a this flurry
of papers recently about using
um the free energy principle in markup
like it's this kind of thing in a very
instrumentalist way
right where where it doesn't mean that
that group that we're
looking at is necessarily
embodying a markov blanket but we can
predict things about that group if we
model it that way
and so if instrumental is reading
for
using active inference as a tool i'm
wondering if
it's going to keep moving in a similar
way that machine learning has where this
is a tool
everybody has a vague concept of what

machine learning is what it does
um and then every company in the world
has their machine learning algorithms to
predict all of their customers behavior
or they want to sell you some machine
learning algorithm about
categorizing some people you're
interacting with or some data you have
um is there
maybe i don't have a specific question
maybe i just wonder what do you see
would what how do you see
active inference just in this like i
guess the social script of how we use
tools and how industries um adopt tools
and then how
just the general public perceives tools
how do you see active inference compared
to its percep the perception of
machine learning wow great
question get it now sorry sorry
i'm so glad you asked the question i
didn't wait like i'm gonna do some
shameless plugging right here
i'm organizing a colloquium on this
exact
question um the equilibrium is going to
be about active inference and collective
intelligence but the real goal is to
discuss
the ethical future of
active inference in terms of social
justice so a
can we not make active inference into
this tool that will mostly serve
capitalistic aims and that will serve to
further oppress communities that have
zero

hermeneutic power now so that's it oh
aim number one
aim number two is to show that active
inference
is this wonderful tool because it's not
opaque it's transparent so we don't have
a black box where it's like
all of these layers of noise you're not
you don't know what they do but it works
so so
with active inference we can see the
parameters of the generated model we can
see how it evolves
even if we're using machine learning to
change the parameters
we can still see what those para
parameters are
so i do see active inference becoming
the future of predictive technology
however i also see that because it's
transparent
and because we are doing this work
of trying to bring together fields that
are thinking about these ethical
questions such as
what is the perspective of a an
oppressed group and how can it inform
our understanding of social dynamics
the beauty of active inference is that
it it
models your anchored position into the
world
this is a fundamental shift because that
that's
intersectionality right there if we can
understand
how intersectionality evolves in the
world how how we can model

a specific person's reality
understanding it through mechanistic
terms
it doesn't have to become this sort of
statistical noise that we have to move
away and
push people into categories you can be
your own phenomenology we do not have to
erase you
and your reality impacts the world in
certain ways so if we wanted to say to
highlight a type of reality a lot of
work is being done with autistic
children
or with gender non-binary like there's a
lot of
of people out there who whose realities
are starting to emerge
because they were erased before through
active inference we could see like
there's this island there
that isn't quite responding like the
rest can we can we zoom in on that can
we see what's happening can we maybe
stop
thinking of society as well you either
fit into the society's mold or you're
you're moved away and
can we think of society as something
that has to adapt to its members
fundamentally thanks for that
amazing thank you for that thanks for
this super interesting
discussion i'll just add a few comments
so one thing is that the
instrumentalist reading which means
using the framework like
a toolkit rather than a depicter about

what the world quote is instrumentalism
and tool building is what allows for a
plurality of narratives to co-exist
including social justice and non-social
justice versions

so because there's a plurality of
narratives that means there's going to
be people who are using the tool who you
disagree with

just like linear models and just like
trains so it's going to be something
that includes a big tent

um rj i'll get to you in a second the
second thing is that the goal of society
is really as you pointed out it's
polysemis there's a lot of features to
it

and it just made me think what is the
optimal skydiving height
and what's the optimal skydiving policy
and for some people

because of cognitive diversity it's
going to be like no just not for me but
no regulation

or not for me not for anyone or there
shouldn't be any regulation and taking
me to the top or i just like two feet
tall

every quadrant is going to exist so
that is the richness of our culture and
i am very hesitant as many people should
be about attempts to quantify goals for
society

and the last point is that what active
inference tools could allow would be
better definition of institutions public
private or

new combinations so for example one

organization could say our affordances
are

this this is our input this is our
output and this is going to be a
currency

arbitrage mechanism that's it we're not
going to be able to censor this other
domain of your life on this other issue
and so modularity and flexibility in
institutions

will become increasingly important and
again that will allow for new
narratives to diversify and that will be
challenging as well as

uh embracing for some rj and then mao
uh i was gonna say two things um so so
one um on uh how much transparency it
offers to

to give us um i think that from from
what i've seen and i'm i'm not an expert
in these areas

but from what i've seen on on its on the
approach toward applying active
inference

in machine learning models um has been
that it offers a lot more auditability
that's that's one thing because the
causal change and then

second uh is what's basically echoing
what dan just said which is that i don't
know that anybody will have any control
over

um other people's use of the tool so i'm
sure it's already being applied in a lot
of these models so i'm sure that it will
end up being

applied to do things that most of us
wouldn't like

but i think that it's commendable that
you're taking efforts to give
people on-ramps toward using it in
positive directions
awesome mao that's exactly it like while
you can't stop a
multi-billion dollar company from using
the system
um you can make it so that people who
usually do not have the means to
partake in the construction of a tool to
become salient voices in the field
one of the things that tends to annoy me
when i go to a conference
is mammals like it seems silly right
but a mantle is just mostly one type of
perspective
and it doesn't to people who are on the
panel it doesn't seem that way to them
they're all different people
with different perspectives but the way
that you came into the world the way
that you approach the research the way
that you approach
paradigms will have some attractive
attracting commonalities because of your
position and
a lot of brilliant work has been done in
shifting paradigms because new
perspectives were coming into the field
intersectionality
probably could only have been possible
because of kimberly crenshaw who came
in in feminism from a very different
point
the second idea is that
we don't have to necessarily control
what the goal state is at every level

but we can probably say that um
there has to be bottom-up and top-down
communication for things to flow more
smoothly right now
highly hierarchical system tend to go
you know top-down and less
bottom-up the communication is a bit
bottlenecked in
one direction and not the other through
active inference we could allow
governance
choose to um allow a better
modeling of what is the optimal goal
state
of ex actor and how can we
model our vision of this local world
to take into consideration all of these
optimal goal states is it possible for
us to optimize
all or a maximum of their goal states um
given where they are and what we have at
our disposition
cool one note there on the bottom up in
the top down and about
including people in the conversation is
that uh
you're really talking about a
reimagining of this nexus
of people and institutions it's a
multi-scale re-understanding
about who and what we are and we're not
going to go into it in this stream but
active inference and free energy has
been applied to sub
organismal things like the brain and
different organs and physiology so we're
not going to really talk about that
but cells are also doing this kind of

thing so it also allows us to look
inwards and more internally and outwards
and more externally
without falling into some of these traps
and then as far as um
how we can make that nexus a better
place the institutions and the power
and the money can be pressured to stay
on the rails and can be uh asked to have
more transparent modeling or say if
you're going to use machine learning on
my preferences then i better
know x y or z or if you're going to
regulate water you better make promises
x y and z
and then also as far as rising up from
the bottom and making the bottom up side
of the communication
you mentioned governance there's also a
lot of cool work with digital governance
related to active inference
and then also just simply including
people in
relationships and conversations and
teams because
that is what the social fabric is so to
not do that
is to not be doing it and to do it is to
be doing it
so that's what i hope would be a cool
way for people to be able to come
together and
talk about through these frameworks and
linear models
though they explain some variants in
some situations you don't get
this kind of conversation with a linear
model so even if it does fall to pure

instrumentalism with active inference
and it's just a way to degenerative
models that explain more variation in
the environment
and future models might explain even
more maybe like a super neural network
that's super next level with a physical
computer
it might explain more variation but if
it can't provide this kind of a
transdisciplinary discussion then it
won't be
as cool as active blue
hi so just to kind of piggyback on like
what everybody's like been
saying so i wonder like active inference
is more transparent right than say like
deep
like neural networks but how is that
like transparency
affected when we don't really know like
at every level
it's like there's this nested hierarchy
and so we don't know
the affordances of all the components at
every level going upward or downward in
the chain of command
and so i just wonder like how we can use
active inference to maybe
like when we don't know like maybe
that's where we find the edges
of the um scripts
right like so like if we're looking at
bottom up or top down is that like
how we probe those edges for
transparency when we don't know
like or without transparency when we
don't know what's happening at all the

different levels

i don't know if that makes sense if that

was just confusing yeah um

just rj one and then i'll first though

the transparent idea that came

from the active um previous one where we

talked about transparent concepts in

philosophy like the ones that are the

water that we swim in so now it's a

different concept with the auditability

of machine learning

but similarly there's this idea about

making the invisible visible

and one suggestion i'd have before rj

them out would be

we aren't just coming to terms on the

inference

remember also that active means that

we're coming to terms on optimal

experimentation

so we might say all right whatever

inference you're making about the world

the danger of some group in

another country what is the policy

and what is the experimentation what's

the info gain we're looking for

we don't need to go for the ultimate

teleos goal of society i want this to be

the case

you can just say okay given the info on

the ground

in this other country what is the policy

that you and i are going to continue

and deepen our relationship over and

that's what's way more important is that

moving forward together

on a journey and on a narrative of

togetherness

rather than coming to agreement on these goals and i get it people want to talk about goals but they're also things that can prevent us from thinking about policy too deeply but justifying it so clearly important so yeah rj and then uh just responding to what blue said uh and it's kind of making me rethink what i said earlier about auditability it sounds like uh from what i've read that it's less getting rid of a black box and more creating a blanket of black boxes so we know where the edges between these black boxes are but that allows us to get a better understanding of what's going on in each of them i suppose so okay wow so yeah so it all depends about in on the granularity that you're using so if you let's say that i started from each individual every person i don't model uh their i don't model their cells or their brain i model them and their behavior say and then you you scale up and up and up and up so the level of granularity of their affordances is going to be obvious at the individual level and the higher you go the more information lost there is which also makes it more computable right like if i if i if i start modeling the

the behavior of a mark of blanket and
then i look at all the other marker
blankets at this level i don't have to
recompute
every individual i have the markup like
it so
it depends on you and your research if
you want to start
by modeling every cell in everybody
and then try to find the emergent
complexity
that's great that's good and you can
have more affordances based on this i
think that's why
scripts are so interesting because they
allow us to
um model the likely affordances
of someone given where they are and
given a set of parameters
so even if it's not their absolute
affordances we can assume that there's
the probability of them
going towards x and now if you start
modeling
all of the possible scripts and how it
connects to
semantic grounding so
what does it mean for something to be a
library what what is
what is the clustering that i can do
then you can start seeing that
some clusterings will start overlapping
in ways that make it more likely for the
person to you know maybe go in this
direction
so for instance um i want to go to the
library
but i live far from the library and i

happen to have a car
well could i fly there
it's unlikely right could i walk there i
mean i
could but i in the scripts of today's
world you tend to want to do things
within
one or two hours if it takes more than
that it's like you're wasting your time
so
i'm probably going to take a car if i
don't have a car i'm probably going to
like either call an
uber because i happen to be in this shoe
in in this um
social demographic or i'll take a bus if
i'm in a different social demographic
you can also predict this based on
whether
you think i care about the environment
so the script of someone caring for the
environment probably won't take a car
right they'll take like the bus or
something so like
you can predict the overlaps based on
uh the kind of scripts that are um
manifest and and measurable awesome
um just a couple comments there as far
as transparency
often people want to know more about the
details of a model and i would just
offer
do we really understand a quadratic
model or addition
or linear regression credit cards
bitcoin
we can use things without quote
understanding them that's a story we

tell
ourselves about the model so i would
offer that
transparency could be approached from
the angle of like
dashboards and really studying the
residuals of your model so if you're
doing an
online learning system and then you find
oh
we had this person in this canyon and
the feedback surveys that we're asking
them
are suggesting they're not getting what
they need so just
you know philosophical debates aside how
are we going to help this person
at this time and perhaps it isn't that
large of an intervention that needs to
be made
in that case so we can separate the
perfect
systems depictions goal from the helping
real people like
in the 100 000 people in this town
there's 500 people who want to go to the
library and will really benefit a lot
how are we going to get these 500 and
then how are we going to make it
open-ended with a survey so that
if people aren't part of our predicted
500 they're still able to contribute
these are things that are at the
intersection of the model
with just some pragmatics and logistics
so i would say the last mile is going to
be just the details of the system and
this

is like the skeleton wow
oh i thought you had a second point i
was waiting for your second point no
that was it okay well no but that let's
see again
this is really beautiful what you said
because we can promote
the resilience of a community
by looking at what is the true goal of
people so like you said an open-ended
survey for these people there's 500 who
want to go to the library
and the rest don't really so but
do they all want to go at x times do
they all live in the same area xyz
knowing this information and knowing
what people actually prefer what would
be their goal state
allows us to enable them through our
resource from the top
to pull down resources that would
actually serve people's purposes
i think that's beautiful social aim that
we could promote through
a new system of allocating resources
depending on people's goals and benefits
yep cool very fun and exciting and
different
active stream so i'll just go to a few
of these
questions many of which we will have
already brought up
in some guys but here we can just put
them all on one page and just
think about what are sort of the
questions that people might have
again either from the acting community
looking out or

from the social sciences or from the
neurosciences and looking in
so how does active inference unify the
disparate
scripts concept so people can raise
their hand but all
emphasize the nestedness and the
hierarchical
like the vertical nesting and then the
heteroarticle the collective behavior
the horizontal the lateral these two
dimensions
and the ways that we can use the
framework instrumentally
to distinguish different models that's a
way that we can
get the strong versus weak scripts
variation axis
and the internalist externalists kind of
together by just emphasizing
the coherence or non-coherence
strongness versus weakness
of different empirical systems and take
a look at them
in this multi-scale way that obviates
the internal external
any if people have thoughts just raise
their hand otherwise i just go
through the oh yeah now go for it that
was a lot of questions
uh the first thing that come that comes
to mind is you you mentioned
a hierarchical hierarchical that's kind
of beautiful so think about it in terms
of
um behaviors that lead to goals right
so imagine that you have these
hierarchies

of goals um that both
are related to time
and to generality so for instance um
i want to be a good person which is like
the highest the largest possible script
ever
because like that's that's one thing
that's like it's kind of binary
it's like okay good bad person whatever
then how do i be a good person well
i can either be a good person in
relation to
um norm expectation or to
values that i myself crafted or values
that were given to me or like
there's like it breaks down and then
within these values set well it breaks
down again into
uh societal behavioral scripts etc so
again see you can see that
as the higher you go the less your
script
is a strong script it's a series of
strong script that becomes this sort of
cluster
now you take your goal at any layer they
pick
any of them your goal is where you want
to get because it happens to be a gate
to get to like
the next stage of your of your script
right
so to get to this goal there's several
possibilities at the lower levels that
lead in this direction
we can thus unify the different uh the
different types of scripts and how we
understanding

through this um sort of pyramid that
goes from elements to goal and we can
cluster which
script doesn't lead to x goal et cetera
et cetera and then we can understand how
um the the individual
models this system has a clear
understanding of the system based on
what they were exposed
to so their environment is what they
were exposed to
their con their their cognition is how
they brought together this sex position
internalized it and started behaving
behaving into the world
um i feel like i answered the question
but it was a very large question
so let me nice that was very
cool and just one other thought there is
that the goal seeking
uh let's not think that that's going to
put our values
in the background because goal seeking
is even more about values
so this is actually not sidestepping the
question of coming to
or stating our values or probable
preferences for
policy this is actually censoring it
and then acting within it so that's the
inference part all right
the next question is in which settings
would an active inference script model
be
valuable and kind of related to that
would be in that setting we can already
ask
what would the metadata for the script

look like and also i'm just curious like
are there any empirical studies that are
being done in the scripts framework or
what

next steps are you taking in this line
of research or what would excite you
out to see like done in a hypothetical
dream experiment

so um the first thing that i think of in
terms of valuable is something that
relates to

let me check his name rj mentioned
we can use

the the the scripts idea in active
inference and the way we modeled it
through weak and
strong to understand

how social behaviors tend to radicalize
or how some policy brought by the top
will be received by the bottom so
anything related to politics or to
social movements would find such a model
valuable the the i'm not sure what you
mean by

metadata so could you be a little more
precise about that

yeah rj why don't you go for that and
maybe give some thoughts what were you
meaning there sure um so first
on um using actin this this

this style model um in political
atmospheres talk about

organizations using active inference in
ways we might not like

um is is governments watching uh
extremist ideology

um so i was i was actually gonna say one
place where i

i wouldn't um i know it will be of
interest to use it but i would love to
see it used this way is to see
how people radicalize um specifically
the scripts we use to react
to uh extremist content um and because
i i think that from my understanding um
that this type of model would be very
useful in identifying when a person has
become
uh what we'll call it like extremist
audiology friendly where they're no
longer going to call it out and say you
know i don't agree with that or i'm not
comfortable
with somebody talking about saying like
domestic terrorism um
so so i could see it being
extraordinarily useful for for
governments in that context
um when it comes to metadata for scripts
like this is included
um so the metadata might be um you know
so
so well the data might describe the
script itself
um the metadata might describe some some
key attributes that
aren't directly related as dan likes to
say the metadata is the data and this is
one of these areas where
like that's definitely there's no the
blur is is definitely large
um so for me when i think about um how i
would
try to start describing the metadata for
scripts it would be
um does this res does this script

require

two people to know that it's happening

so like for example um uh flirting

it might be you know um one person can

enact their scripts without the other

person uh

enacting it whereas like a script for

swearing in the president of the united

states

that is a script that that's a strong

script with explicit rules that

everybody has to be aware of otherwise

it doesn't work

um so i i think that there's like

categorization

values that might be associated with it

and then uh yeah but anyway i think

there's a lot you could go into in that

so i won't go

too far down the rabbit hole cool let's

do blue then

shannon then mao

so just to piggyback off of what rj was

saying about

radicalization it's interesting like

when did radicalization

become a script right and like this is

what we're talking about finding the

edges

and so like where does the script even

exist is it like an external construct

like a societal construct this is the

battle between internalism and

externalism

but when is there like a critical mass

of

internal scripts that develop that then

it becomes this external script like you

can be a radical
like the first person to become
radicalized like was just
radical right like they just took their
own initiative
to to you know it was their own script
it was like an internal construct
and so where's the like the line really
like when a script becomes
an acceptable or even an alternative way
like there's some kind of critical mass
that has to accrue before before that
happens and it's just
something interesting to think about
shannon

then mo and then rj again i have a bit
of a topic change so maybe you want to
talk about that first and then ask me a
question

okay if it's on the same um idea mao
then rj

yeah so exactly so how do we figure out
when someone is going to radicalize or
specifically

what leads to radicalization well the
truth is radicalization

tends to happen within clustering
so it's not just someone who unless this
person has

specific like um
psychological issues which make them go
off on their own tangent
away from the cluster it tends to be
that

clusters form around narratives right so
what is

let's say a meninist group what is a
meninist narrative well women are

gaining too many rights and they're
taking the spot of men and in some men
in his group
this narrative becomes women are trying
to take down men
to enslave them and then the narrative
moves forward and it moves
women are trying to destroy men and then
what is the acceptable answer to this
well we must destroy women
and this is something you can see
happening through the shift
of the narrative the the people inside
these clusters
start bringing categories together and
then pinging off each other
and then i'm giving you this narrative
and a lot of people
oh yeah you're right this is the right
narrative and then somebody else with
slightly different a little bit more
narrative keeps pushing on and you get
this
cluster and then people build on each
other and the person on top or at least
someone who's believing
so much into this narrative which also
identifies it with potentially violence
which
we could measure based on the type of
narrative that's being used
we will possibly be able to identify
okay well this cluster of people
and specifically this type of person in
there is is more likely to do something
really bad really soon what that means
for our governments
that's not up for me to decide ideally i

think it would be
um bringing help in this in these areas
and being like well
maybe maybe you need to rethink your
narratives maybe we can help you soften
whatever it is you're going through
that's
pushing you to to get towards these
narratives
thanks rj
uh well now covered a lot of what i was
going to say um once again
perfect we're definitely synced um i
would just add that it is definitely one
of those areas
where um uh i think
we people who are who are working in the
topic outside
of governments um to
try to create the speed bumps
um so that like like that not not i
don't want to say speed bumps it's hard
to articulate like like
the the like clearly express the
limitations so that
um government doesn't go and take this
and then apply it badly like they've
done
with um with machine learning and say
like the judicial system
uh we're trying to predict flight risk
and things like that they applied it
very quickly
and didn't really weren't really sure
what they were doing and they applied it
before they really understood what they
were working with and
obviously did a lot of damage so thanks

just one last comment on this
uh radical doesn't mean divergent or
committed
radical means from the root so i think
that radicalization as a discussion
should refocus us on the seed memes the
germ memes in the space
and the ways that generative models
modelable by script and active inference
could understand what leads to the
adoption the phase transition of the
seed memes
and then the policy decisions that
someone engages in are probabilistic
and can be inferred downstream of these
seed memetics
and so those are the premises of the
narrative so if someone thinks that
they're in braveheart
or if they think that they're in batman
or if they think that they're a
different role in a movie
that's really what we're getting at is
what is the genre
of culture what kind of movie is it a
good one short one long one
what is the cast going to be how are
they going to treat each other
we can talk about it that way and that
is what has been distorted in the public
eye
for too long shannon
and then sasha thanks that was um
a really informative discussion and that
metaphor really brought it all together
at the end daniel
um so i don't study social
interactions on that large of a scale i

bring people in a lab i look at how
maybe their
motor activity coordinates in in like
postural sway so something that you're
not really aware of happening but it's
happening all the time
you're swaying and maybe you're
mirroring someone um across from you
and you guys have been having a
conversation of what the metadata
looks like um but what is i want what
does the
data look like if you were going to use
this active inference model
at something as large scale as these
social movements that you're talking
about
um or at smaller scales like i i can see
that
the kind of data i would look at in very
small groups
would be you know patterns of motor
activity
maybe that i'm measuring from filming
the body
or motor activity in a brain um but in a
large
social group like that would you be
looking at like gps tracking a phone
like we've been doing a lot of
modeling of how people interact in this
um pandemic
and where large clusters of people are
interacting and
we might have like super spreading
events for covid19
um or would these narratives that you've
been talking about um

sound like their verbal narrative so
would you be looking at um
social media profiles um you know
twitter facebook
and how
honest are those data so i know that the
data i'm measuring the
the postural sway data my participant i
mean unless they're actually
swaying back and forth um intentionally
like to beat music they don't have
control over
all of these small muscle movements in
your body that are
that are controlling your posture um
it's really a very embodied
behavior and i know that when i'm
measuring it i'm getting an honest
signal
but if you're looking at um even gps
data or if you're looking at
social media narrative data the signal
could
like people could be intentionally
signaling for
others who share their view or others
who don't share their view or
intentionally not signaling because
they're afraid of
a response or all of the various things
that could
affect and make the make the signal that
you're measuring not be honest so
so what would the data look like in your
that that would make your active
inference model of this
large social situation be
closer to the truth thanks for that

awesome question we're gonna do sasha

then then

rj yeah i guess continuing the

conversation of

how um i would like to see this used

um thinking about scripts and

stereotypes

um i guess my my hope for seeing active

inference as

a tool is for people to um

think about it think about their own

role in

their active inference story um and that

instead of just thinking about machine

learning and like um getting

recommendations and ads based on your

preferences

that you actually um shape your own

story

and that you are building your narrative

which then leads to policies and goals

and so um taking a more active role

in the kinds of things that will be

predicted about us based on our actions

um and to try to shape that in a way

that signals um

something deeper uh like our values

ooh mao and on the data question two

yeah so um i'm gonna try and uh if i

forget a question they were important i

really don't wanna so if i forget please

remind me

the first question was what kind of data

well you're right social media would be

super great because what you said about

the honesty of data is that well

social reality is manifest reality like

if you behave a certain way because you

know your group behaves a certain way
well it's because you've integrated what
the group wants and that's the script
the the way we could see whether or not
this is something that your cognition is
entirely geared towards or is just
fulfilling because xyz
is how you behave in other groups so
one of the things that would be
difficult to highlight
is if you are let's say um part of a
cult right
you're part of a cult and you are of no
contact with anyone else
it would be difficult for us to figure
out whether this type of behavior
is also being contested inside we could
maybe
model this by looking at all of the
other members and
what kinds of interactions some people
who tend to disagree a little bit
tend to have we can't know for sure and
that's kind of also
that's we make inferences about the
states of our agents
based on what they put out into the
world the observations they put out
you're right that social media data
would not necessarily be enough though
because
um you want to have data about what it
means to be in the world even outside of
the virtual world
so if all you care about is online
behavior
then well i mean it doesn't really
matter how you move but

if what you care about is scripts as a whole in society well we would use um narrative data so what are the movies how do people in the movies behave and talk what is literature what are the words that are being used in what way with what valence is the positive or negative then you would use um your rights data about how people behave together to do experiments by connecting people and doing hyper scanning data like there's all of the data that would allow us to understand how a specific semantic cluster tends to be grouped and then tends to be connected causally all of this would allow us to derive scripts it all depends about on where you want to apply this knowledge what was the second question i think that covered i think i just sort of reworded my original question but that was great i think sasha had a question and i forgot sasha yes sasha yeah not so much a question just a statement for how can people um think about applying scripts and active inference to their own behavior and behave in ways that signal at our true goals in ways that are script and socially acceptable

okay okay so um this is the really cool
part
of having a model that
can take both higher level and lower
levels and
you're right that giving people power
over their data
would also help us figure out what their
true goal is
right so if you if you're like i want to
take an active part
in shaping my reality well essentially
you're giving us what your true goal
is so that's good for everyone so you're
right
um it would be great for society if
people had this control
i think that's where we're trying to get
towards in ethical circles where we're
like
we're not just going to take stuff from
you and then
model and then you won't really have any
benefit from it or if you have a benefit
we define what that benefit is it's more
like
we can show you the data that is yours
we
can show you the data that is aggregated
from you as well
and you get to decide what kind of
person you want to be based on this so
we show you the affordance scape around
you based on your profile
well now you can pick what you want to
do with this affordance escape
cool and just so interesting about the
inference

and then moving towards action and not just targeting the ads or just targeting the consumption but actually targeting the participation as well

and then we'll speed through the figures

oh sure uh i would just add that this is definitely an area where it feels like the

the data and the metadata of the descriptive data about the data

uh definitely blur and also uh shannon you were talking about

people not having control over how they sway et cetera and i just wanted to thank you for making me hyper aware of my own posture

nice um cool well what a fun

discussion these are the figures and i think for next week people can start to think about what questions they want to address

and then um for now i would just denote we should talk about the figures and about how they relate to some of these more qualitative things we discussed today like maybe we could even work through one of these diagrams for flirting or for ordering a cup of coffee

um figure two was a layout like a graphical layout of the internal and the external realizer

and so we could ask where's the markov blanket here and what does that say about how the world is what does that say about how we could model the world

what are the points where we could
intervene what are the experiments that
could be informative
whose perspective could we include when
we're thinking about this
like all these big questions that are
what should be at the front of our mind
when we approach a system like this
so that was pretty cool figure let's
also come back to that
and uh that's basically it in our last
couple minutes we'll hear a final
note so anyone raise your hand if you
want to make a short note
other than that though we will give a
form for a survey
and especially for 10.2 questions from
everyone who's participating
if you're watching it live or in replay
just
add a comment with something we could
address in 10.2
and uh stay in communication and other
than that
let's hear some last thoughts or what
was your favorite part of the discussion
from anyone who wants to talk
um rj then mao um
i after after this doug i'm definitely
all the more interested
in what kind of threshold events there
are for
the creation of scripts and for the
breaking of them
awesome okay
um i'm super excited to see that people
are interested because
when i started thinking about this and

and and started working on it with
maxwell
my thought was i'm going to be this
fringe researcher that no one ever
quotes because no one cares
and it has the beautiful side that no
one will ever criticize what i do
but but i'm really excited to see that
people are being um inspired by this
work people start
asking questions and they bring in new
possibilities um people from the field
of music are interested which means that
there's gonna be uh way more input from
a variety of fields which is going to
make this
super rich and i'm really thankful for
your um
i'm not your administration um
you you led this discussion really
really well and that was very pleasant
to be part of thank you
action stream has the best norms and the
best participants
i have to say no i'm just kidding but
really thanks everyone
for making the discussion so excellent
because that's really what it is
and uh we hope to see you again in 10.2
hear anyone else's thoughts see you in
future active streams so
thanks and we will talk to you next time